

IT-Based Landscape Stratification for Rain-Gauge Network Design

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1. Introduction

Designing an optimal precipitation-monitoring network is not a simple geostatistical optimisation task. It requires understanding:

- how **landscape structure** affects rainfall organisation,
- how **information gain** varies across space,
- how **hydrological response** depends on spatial rainfall patterns.

This document implements **Layer 1** of a modern three-layer network-design framework:

1. **Physiographic stratification** (structural representativeness)
2. **Information-gain optimisation** (analytical efficiency)
3. **Hydrological coupling** (functional adequacy)

Here we implement **Layer 1**: an objective stratification of the Burgwald landscape using **information-theoretic metrics** and **pattern-based texture signatures**.

2. Conceptual background

2.1 Physiographic stratification

Goal: ensure that rainfall stations represent the dominant structural units of the landscape. Landscape units differ in:

- land-cover composition
- forest–openland transitions
- patch fragmentation and clumpiness
- canopy interception potential
- micro-topographic exposure

We capture these differences using information-theoretic metrics:

- **Entropy $H(x)$**
 - measures thematic diversity of land cover
 - low $\rightarrow$ homogeneous (e.g. closed forest)
 - high $\rightarrow$ heterogeneous, mixed, patchy
- **Relative mutual information U**
 - measures configurational order (clumpiness)
 - low $\rightarrow$ fragmented patterns
 - high $\rightarrow$ ordered, aggregated patterns

Each 2×2 km grid cell gets a structural fingerprint in **(H, U)-space**, and clustering these fingerprints yields **objective structural strata**.

2.2 Information-gain optimisation

Once strata exist, statistical optimisation (e.g. kriging variance, radar–gauge mismatch, representativeness error) should be applied **within and across strata**, not across the whole catchment at once. This avoids bias towards “statistically convenient” open areas and keeps the optimisation linked to process-relevant structural units.

2.3 Hydrological coupling

Finally, a network must improve the prediction of:

- streamflow peaks (P–Q coherence)
- storage changes
- water-balance residuals

Even a statistically optimal station is useless if it does not represent rainfall regimes controlling the hydrograph. This QMD prepares the structural layer required for such coupling.

3. Load AOI and land cover

```
# Load AOI and land cover
lc_burgwald <- terra::rast(clc_file)
aoi_burgwald <- sf::read_sf(aoi_file)

# Harmonise CRS
if (!sf::st_crs(aoi_burgwald) == sf::st_crs(lc_burgwald)) {
  aoi_burgwald <- sf::st_transform(aoi_burgwald, sf::st_crs(lc_burgwald))
}

# Clip raster strictly to AOI
lc_burgwald <- lc_burgwald |>
  terra::crop(terra::vect(aoi_burgwald)) |>
  terra::mask(terra::vect(aoi_burgwald))

# Ensure integer categories for landscapemetrics
terra::values(lc_burgwald) <- round(terra::values(lc_burgwald))
```

4. Compute IT metrics on a 2×2 km structural grid

Concept:

- Create a regular **2×2 km** grid over the Burgwald.
- Treat each grid cell as a small “landscape”.
- Compute **entropy $H(x)$** and **relative mutual information U** for each grid cell.
- Each cell becomes a point in (H, U)-space.

```
# Transform AOI to metric CRS so that cellsize is in metres
aoi_utm <- sf::st_transform(aoi_burgwald, 25832) # ETRS89 / UTM 32N

# Grid resolution (metres)
grid_cellsize <- 2000 # 2 km × 2 km

# Build grid in UTM space
grid_utm <- sf::st_make_grid(
  aoi_utm,
  cellsize = grid_cellsize,
  what      = "polygons"
) |>
  sf::st_as_sf() |>
  dplyr::mutate(grid_id = dplyr::row_number())

# Back-transform grid to raster CRS
grid <- sf::st_transform(grid_utm, sf::st_crs(lc_burgwald))

# Compute entropy and relative mutual information per grid cell
lsm_grid_it <- sample_lsm(
  landscape = lc_burgwald,
  y         = grid,
  what      = c("lsm_l_ent", "lsm_l_relmutf"),
  level     = "landscape"
)

# Wide format: grid_id | ent | relmutf
lsm_grid_it_wide <- lsm_grid_it |>
  dplyr::select(grid_id = plot_id, metric, value) |>
  tidyr::pivot_wider(names_from = metric, values_from = value)

grid_it_sf <- grid |>
  dplyr::left_join(lsm_grid_it_wide, by = "grid_id")
```

5. Cluster cells into IT-based strata

Concept:

- Use (H, U) to cluster grid cells into **k structural strata**.
- Each stratum is a region with similar composition and configuration.
- We will later select representative cells per stratum as station candidates.

```
# Extract numeric metrics
it_clust_df <- grid_it_sf |>
  sf::st_drop_geometry() |>
  dplyr::select(grid_id, ent, relmutinf) |>
  dplyr::filter(!is.na(ent), !is.na(relmutinf))

# Standardise to z-scores
it_clust_scaled <- it_clust_df |>
  dplyr::mutate(
    ent_z = as.numeric(scale(ent)),
    relmutinf_z = as.numeric(scale(relmutinf))
  )

# Number of strata
k_strata <- 5

set.seed(123)
km_it <- stats::kmeans(
  it_clust_scaled[, c("ent_z", "relmutinf_z")],
  centers = k_strata,
  nstart = 50
)

it_clust_scaled$stratum_id <- km_it$cluster

# Attach strata to grid sf
grid_it_strata_sf <- grid_it_sf |>
  dplyr::left_join(
    it_clust_scaled |> dplyr::select(grid_id, stratum_id),
    by = "grid_id"
  )
```

6. Select multiple representative cells per stratum

We now choose several “typical” cells per stratum (here: 4 each → 20 candidates total):

```
n_per_stratum <- 4

# Stratum centroids in z-space
strata_centroids <- it_clust_scaled |>
  dplyr::group_by(stratum_id) |>
  dplyr::summarise(
    ent_z_mean      = mean(ent_z, na.rm = TRUE),
    relmutinf_z_mean = mean(relmutinf_z, na.rm = TRUE),
    .groups = "drop"
  )

# Distance to stratum centroid
it_clust_scaled <- it_clust_scaled |>
  dplyr::left_join(strata_centroids, by = "stratum_id") |>
  dplyr::mutate(
    dist_to_center = sqrt(
      (ent_z - ent_z_mean)^2 +
      (relmutinf_z - relmutinf_z_mean)^2
    )
  )

# n nearest cells per stratum
rep_cells <- it_clust_scaled |>
  dplyr::group_by(stratum_id) |>
  dplyr::slice_min(order_by = dist_to_center,
                   n = n_per_stratum,
                   with_ties = FALSE) |>
  dplyr::ungroup()

# Attach geometries
rep_cells_sf <- grid_it_strata_sf |>
  dplyr::inner_join(
    rep_cells |>
      dplyr::select(
        grid_id,
        stratum_id,
        ent,
        relmutinf,

```

```

    ent_z,
    relmutinf_z
  ),
  by = c("grid_id", "stratum_id")
)

```

7. Compute station-candidate centroids

```

station_candidates_sf <- rep_cells_sf |>
  sf::st_centroid() |>
  dplyr::mutate(
    station_id = dplyr::row_number()
  ) |>
  dplyr::select(
    station_id,
    stratum_id,
    grid_id,
    ent,
    relmutinf,
    ent_z,
    relmutinf_z
    # geometry is kept implicitly by sf
  )

```

8. Pattern-based signatures (motif COVE)

Concept:

- motif calculates **co-occurrence signatures** on moving windows.
- Here: window size 25×25 cells (~2.5 km at 100 m resolution).
- Each window yields a probability distribution of adjacency patterns.
- Distances between signatures (e.g. Jensen-Shannon) define **pattern types**.

```

lsp_cove_burgwald <- lsp_signature(
  x          = lc_burgwald,
  type       = "cove",
  window     = 25,
  normalization = "pdf"
)

```

```

)

dist_mat <- lsp_to_dist(
  x      = lsp_cove_burgwald,
  dist_fun = "jensen-shannon"
)

k_pattern <- 6
set.seed(123)
hc <- hclust(dist_mat, method = "ward.D2")
pattern_ids <- cutree(hc, k = k_pattern)

lsp_cove_burgwald$pattern_cluster <- pattern_ids

lsp_pattern_sf <- lsp_add_sf(lsp_cove_burgwald)

# Reduce list-columns for plotting / joins
geom_col <- attr(lsp_pattern_sf, "sf_column")
lsp_pattern_sf_plot <- lsp_pattern_sf[, c("pattern_cluster", geom_col)]
lsp_pattern_sf_plot$pattern_cluster <- as.factor(lsp_pattern_sf_plot$pattern_cluster)

```

9. Attach pattern types to station candidates

```

station_with_pattern_sf <- sf::st_join(
  station_candidates_sf,
  lsp_pattern_sf_plot,
  join = sf::st_intersects,
  left = TRUE
)

```

10. Visualisation (optional)

```

mapview(grid_it_strata_sf, zcol = "stratum_id")
mapview(station_with_pattern_sf, zcol = "stratum_id", cex = 4)
mapview(lsp_pattern_sf_plot, zcol = "pattern_cluster")

```


11. Outlook: combining IT-strata with classical physiographic stratification

The IT-based stratification identifies **landscape structural regimes**, but it should be combined with more traditional physiographic factors.

Future integration should combine IT strata with:

- **Elevation bands** → capture orographic rainfall gradients
- **Slope and aspect classes** → control exposure, interception, wind undercatch
- **Geological units / parent material** → influence stormflow generation processes
- **Hydrotopes / representative hillslopes** → link rainfall structure to runoff response
- **Forest structural types** (deciduous, coniferous, mixed, stand age) → modify canopy storage and interception patterns
- **Cold-air drainage pathways** → relevant for radiation fog and winter precipitation phase transitions
- **Convective storm tracks / radar climatology** → areas repeatedly hit by convective cells
- **Accessibility and operational constraints** → maintenance, vandalism, telemetry, power supply

The final Burgwald network should merge:

1. **IT strata** – structural representativeness
2. **Physiographic strata** – process-based representativeness
3. **Information-gain optimisation** – cost-efficient densification
4. **Hydrological validation** – functional adequacy

Together, these layers define a scientifically defensible, resource-efficient, and hydrologically meaningful precipitation-monitoring network for the Burgwald.